Xiongxin Yang

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Research Interests

Discrete probability; Markov chain Monte Carlo method, Markov random field; statistical and learning theory; computational and discrete geometry; mathematics for AI, AI for mathematics.

Education

University of California, Santa Barbara — Sept 2025 – Present
Ph.D. Student in Computer Science
- Advisor: Prof. Eric Vigoda

Northeast Normal University — Sept 2020 – Jun 2024
B.S. in Computer Science and Technology
- Advisor: Prof. Zhiguo Fu

Employment

The University of Hong Kong — May 2025 – Jul 2025
Research Assistant hosted by Prof. Weiming Feng

New York University Shanghai — Oct 2024 – Jan 2025
Research Associate hosted by Prof. Jie Xue

Visits and Internships

The University of Hong Kong — Mar 2025
Visitor hosted by Prof. Weiming Feng

Nanjing University — Apr 2023
Visiting Student hosted by Prof. Yitong Yin

Selected Awards and Scholarships

Presidential Scholarship (Northeast Normal University) — 2022 & 2024
Ranked 1st in the major

National Scholarship — 2021
Top national scholarship in mainland China (top 0.2% nationwide)

Publications and Preprints

Unless noted otherwise, authors are listed alphabetically, following standard practice in theoretical computer science. ● denotes conference publications; ● denotes journal publications.

[8] Exponential-Time Approximation Schemes for Ordering Problem
Huairui Chu, Ajaykrishnan E S, Viktoria Korchemna, Daniel Lokshtanov, Anikait Mundhra, Saket Saurabh, Thomas Schiber, Xiaoyang Xu, Jie Xue, **X. Yang**.
In progress

[7] Simple and Efficient Algorithms for Approximating Maximum Depth of Unit Disks and Unit Balls
Sujoy Bhore, Jie Xue, **X. Yang**, Jiumu Zhu
Submitted

[6] Fully Dynamic Diameter Approximation in Unit Disk Graphs
Sujoy Bhore, Jie Xue, **X. Yang**, Jiumu Zhu
Submitted

[5] Near-Optimal Dynamic Data Structures for Maximum Depth and Klee's Measure of Boxes
Sujoy Bhore, Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
53rd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2026**)

[4] Learning CNF formulas from uniform random solutions in the local lemma regime
Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
58th Annual ACM Symposium on Theory of Computing (**STOC 2026**)

[3] Dynamic maximum depth of geometric objects
Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
41st International Symposium on Computational Geometry (**SoCG 2025**)

[2] Spectral Independence Beyond Total Influence on Trees and Related Graphs
Xiaoyu Chen, **X. Yang**, Yitong Yin, Xinyuan Zhang
2025 ACM-SIAM Symposium on Discrete Algorithms (**SODA 2025**)

[1] Beyond windability: Approximability of the four-vertex model
Tianyu Liu, **X. Yang**
Theoretical Computer Science, Volume 995 (2024), Article 114491

Talks

- Dynamic Maximum Depth of Geometric Objects
  - The 41st International Symposium on Computational Geometry (SoCG 2025)
    Jun 25, 2025 @ Kanazawa, Ishikawa Prefecture, Japan
  - The 16th Annual Meeting of Asian Association for Algorithms and Computation (AAAC 2025)
    May 31, 2025 @ The City University of Hong Kong, Hong Kong SAR, China
- A Glimpse into the Theory World
  - Information Science Academic Forum (in Chinese)
    Apr 11, 2025 @ Northeast Normal University, Changchun, China
- Approximability of the Four-Vertex Model
  - NYUSH Theory Lunch
    Oct 21, 2024 @ New York University Shanghai, Shanghai, China

Grants

Catalyst Grant Program (Cohere Labs) 2026
$1000 API credit

Academic Service

Conference Reviewing: SoCG (2025), WWW (2026)